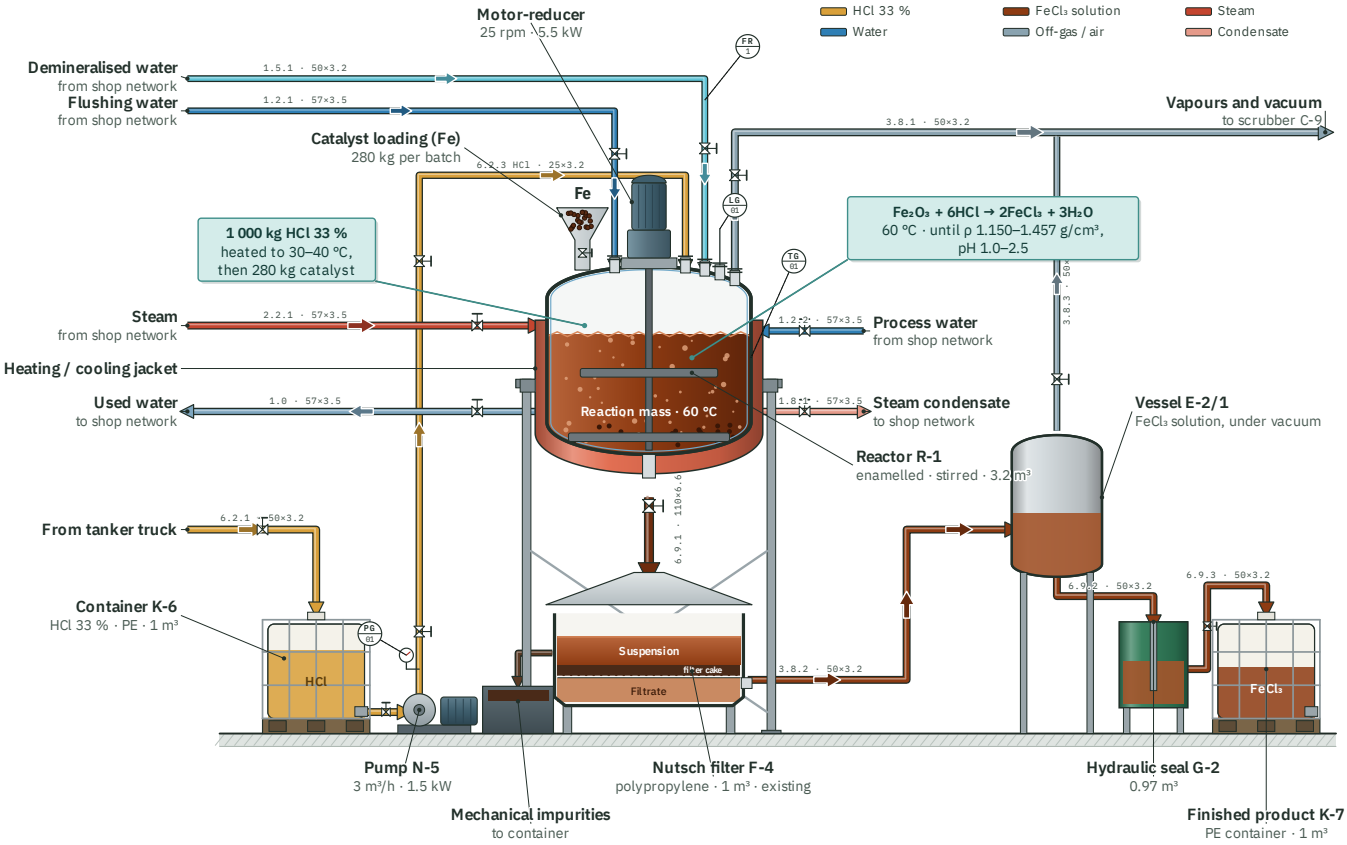
# IRON.

$$Fe_2O_3 + 6HCl \rightarrow 2FeCl_3 + 3H_2O$$

SINGLE-STAGE REACTION · 60 °C

## Iron(III) chloride ($FeCl_3$) produced from spent industrial catalyst

A compact single-line unit that dissolves the spent iron-oxide catalyst of an ammonia plant's first-stage CO converter in hydrochloric acid — turning an industrial waste into a coagulant reagent for thermal power plants and the chemical industry.




**500** t / year
Design output of $FeCl_3$ solution

**280** kg
Spent catalyst reused per 1 t of product

**60** °C
Reaction temperature in a steam-jacketed reactor

**1** line
Batch process, four operations

**GOST** 4147-74
Product quality standard

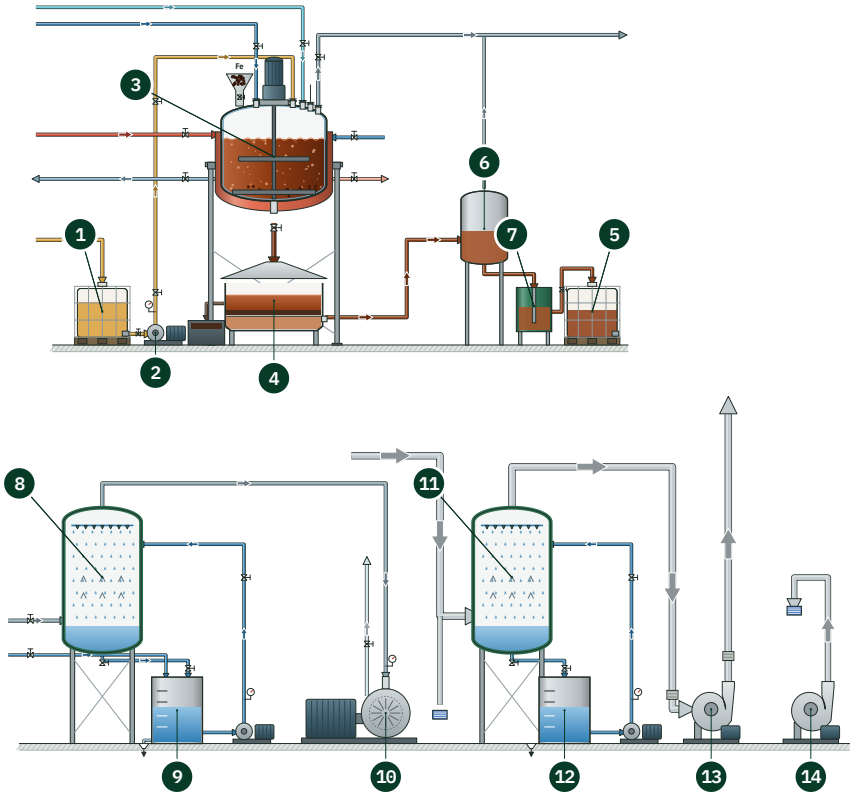
**Principle.** Iron in the spent catalyst is already trivalent ($Fe_2O_3$, 70–75 %), so it converts directly to $FeCl_3$ — no oxidation stage, no nitric acid.

| | |
|---|---|
| Feedstock | **Spent CO-converter catalyst** |
| Reagent | **Hydrochloric acid 33 %** |
| Product | **$FeCl_3$ solution, dark brown** |
| Shelf life | **12 months** |

THE INSTALLATION

# One line. Four operations. Two-stage gas cleaning.

Every item shown is taken from the unit's process & instrumentation diagram and equipment specification. Reactor off-gases and extracted air pass through two scrubbers before cleaned air is released.




HCl 33 % · $FeCl_3$ solution · Steam · Water · Off-gas / air

Section drawing based on the process diagram · not to scale

| | | | |
|---|---|---|---|
| 1 **K-6 HCl feed container** Polyethylene · 1 m³ | 2 **N-5 HCl transfer pump** 3 m³/h · 1.5 kW | 3 **R-1 Jacketed reactor** Enamelled · 3.2 m³ · 5.5 kW | 4 **F-4 Nutsch filter** Polypropylene · 1 m³ |
| 5 **K-7 Product container** Polyethylene · 1 m³ | 6 **E-2 Vessel** 0.97 m³ · Ø920 mm | 7 **G-2 Hydraulic seal** 0.97 m³ | 8 **C-9 Scrubber 1** Rubber-lined · 5.6 m³ |
| 9 **E-9 Water tank** 1.5 m³ · pumps 3 m³/h | 10 **VN-3 Vacuum pump** 240 m³/h · 5.5 kW | 11 **C-8 Scrubber 2** Rubber-lined · 5.6 m³ | 12 **E-8 Water tank** 1.5 m³ |
| 13 **VV-2 Exhaust fan** 5 600 m³/h · 1.5 kW | 14 **VP-1 Supply-air fan** 9 600 m³/h · 1.5 kW | | |

TECHNOLOGICAL PROCESS · BATCH

01
## Dissolve
**1 000 kg** HCl 33 % pumped into R-1 and heated to **30–40 °C** by the steam jacket; **280 kg** of crushed catalyst added gradually.

02
## React
Solution brought to **60 °C**; density and pH analysed until **ρ 1.150–1.457 g/cm³** and **pH 1.0–2.5**.

03
## Filter
Nutsch filter F-4 retains the insoluble residue; the $FeCl_3$ solution is separated.

04
## Package
Sealed polyethylene containers, stored at **20–40 °C**, protected from sun and precipitation.

GAS CLEANING — Reactor off-gas → scrubber C-9 + Extracted air → scrubber C-8, fan VV-2 → Cleaned air to atmosphere

### Built into an operating site
Reactor located in building 361/1 of the Soda shop; the catalyst comes from the site's Ammonia-3 shop.

### Existing utilities
Demineralised water, process water and steam are taken from the shop networks; condensate is returned.

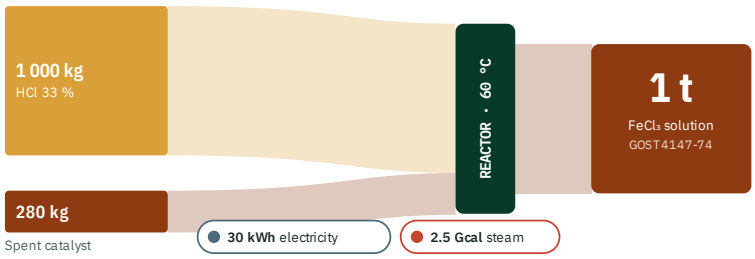
### Energy demand
30 kWh of electricity and 2.5 Gcal of steam per tonne of product; installed drives ≈18.5 kW in total.

RAW MATERIAL

# A waste stream becomes the main raw material

Spent catalyst replaces purchased iron feedstock. Below are the project's temporary consumption norms per tonne of $FeCl_3$ solution; annual values are these norms multiplied by the 500 t/yr design capacity.

## Resource balance · per 1 t of product




## At design capacity · per year

| | |
|---|---|
| **500 t** $FeCl_3$ solution output | **140 t** Spent catalyst turned into feedstock |
| **500 t** Hydrochloric acid 33 % required | **15 MWh** Electricity · plus 1 250 Gcal steam |

Calculated: consumption norm × 500 t/yr

## Route comparison · how $FeCl_3$ is made

| | | |
|---|---|---|
| **From scrap metal**<br>Fe(II) must be oxidised to Fe(III) | Scrap + HCl → $FeCl_2$ → + $HNO_3$ + extra HCl → $FeCl_3$ | Multi-stage, labour-intensive |
| **From spent pickling liquor**<br>Low $FeCl_2$ content, Ca & Na chloride salts | Pickling residue → Excess $HNO_3$ + HCl → $FeCl_3$ | Multi-stage, labour-intensive |
| **IRON · spent catalyst**<br>Iron already trivalent ($Fe_2O_3$ 70–75 %) | Catalyst + HCl 33 % → $FeCl_3$ | **Direct, single stage, fast dissolution** |

Source: developer's technical note on existing $FeCl_3$ production methods

## Product quality · GOST 4147-74

| | | |
|---|---|---|
| Appearance | | Dark-brown liquid, readily water-soluble |
| $FeCl_3 \cdot 6H_2O$ content | **42–55 %** | 0 — 100 % |
| Density at 20 °C | **1.150–1.457** | 1 — 1.6 g/cm³ |
| pH | **1.0–2.5** | 0 — 7 |
| Water-insoluble residue | **0.05 %** | mass fraction |
| Guaranteed shelf life | **12 months** | sealed PE containers, 20–40 °C |

## Where the product goes

**Thermal power plants**
Reagent (coagulant)

**Chemical industry**
Process reagent

## Features

- **Direct conversion:** Fe(III) becomes $FeCl_3$ in one stage
- **No nitric acid:** HCl is the only reagent
- **Co-coagulants:** Cu & Cr chlorides speed up coagulation
- **Waste as feedstock:** spent catalyst replaces purchased iron

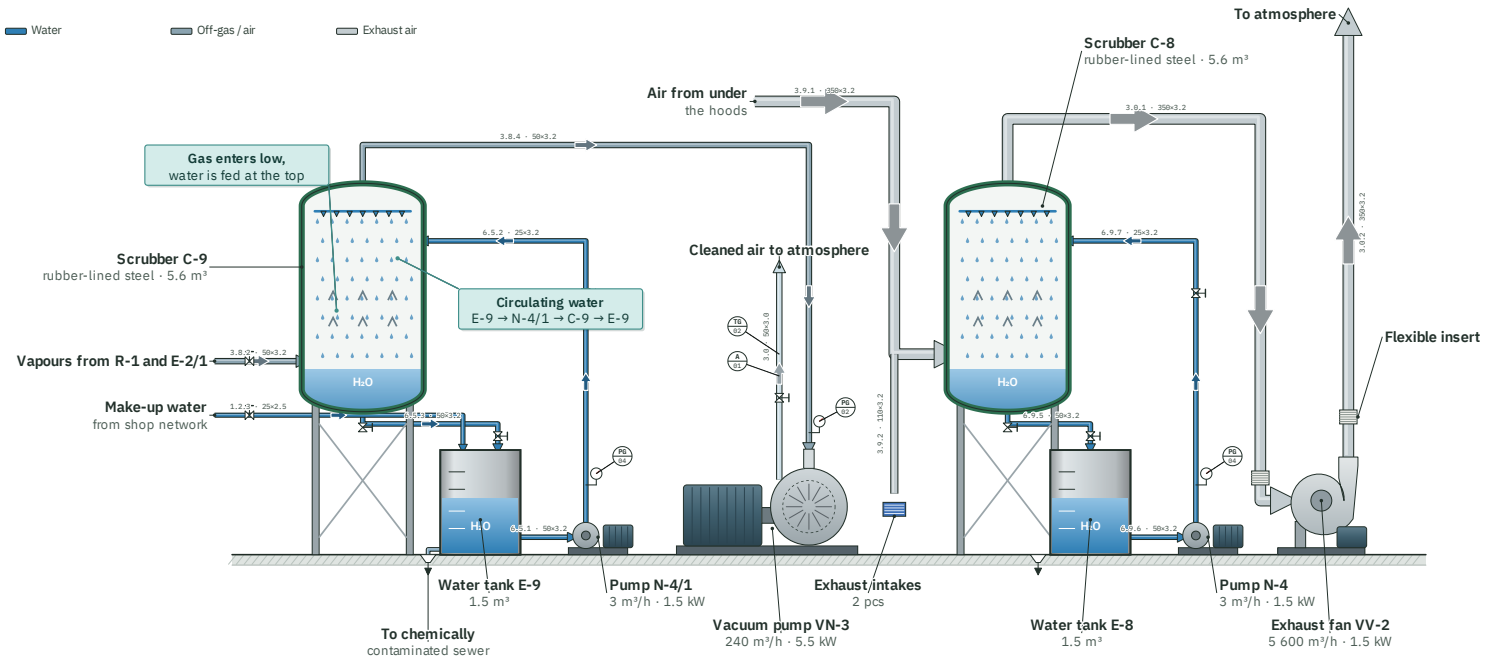
RAW MATERIAL SPECIFICATION

**Hydrochloric acid · GOST 3118-77**
colourless liquid · HCl not less than 33 % · 1.19 g/cm³ · pH 1

**Spent catalyst · CO converter stage 1, Ammonia-3 shop**
dark-brown granules · $Fe_2O_3$ 70–75 %

GAS CLEANING

# Two-stage scrubbing of off-gases




Section drawing based on the process diagram · not to scale

IMPLEMENTATION ROADMAP

## 2021
### Initiative
Working documentation based on the Soda shop head's memo of 28.10.2021.

## Design
### Engineering package
P&ID, equipment specification, technological description and consumption norms.

## Installation
### Equipment
Reactor, scrubbers, pumps and fans to be installed; the nutsch filter already exists on site.

## 2024
### Pilot-industrial unit
Implementation term of the 500 t/yr line set in the technical description.

DESIGN INDICATORS

**500** t/yr
$FeCl_3$ coagulant solution to GOST 4147-74

**≈140** t/yr
spent catalyst put back into use as feedstock

**2** scrubbers
clean the off-gases before release to atmosphere

FINISHED PRODUCT

# Transport and storage

01 Transported by any vehicle in line with the transport rules

02 Sealed polyethylene containers, 20–40 °C, protected from sunlight and precipitation

03 Guaranteed shelf life 12 months; afterwards usable once re-checked against the standard

**Hazardous Waste Management Directorate**
under the Agency for Waste Management and Circular Economy Development

PHONE *TO BE ADDED*
E-MAIL *TO BE ADDED*
WEB *TO BE ADDED*